# BiTempQA: A Diagnostic Benchmark for Bitemporal Reasoning in LLM Agent Memory Systems

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## Abstract

Memory-augmented LLM agents must track not only *what* happened but *when* it happened and *when they learned about it*, a dual-timestamp (bitemporal) distinction critical for real-world applications such as medical records, financial systems, and news analysis. Existing benchmarks either ignore timestamps entirely or treat time as a single dimension, leaving a gap in evaluating this capability.

We introduce **BiTempQA**, a diagnostic benchmark of 1,536 Chinese question-answer pairs (308 held-out test) across 10 scenario types and 9 question types at three difficulty levels, where each memory entry carries explicit `event_time` (when something occurred) and `record_time` (when the system stored it) annotations. Over half (56.5%) of questions require reasoning about both timestamps simultaneously. We evaluate five memory backends (FAISS, BM25, Naive RAG, Simple KG, ChromaDB) through a unified Retrieve→LLM Generate→LLM Judge pipeline. A mixed-effects logistic regression confirms that record-time reasoning is genuinely harder than event-time reasoning ($p$=0.023), surviving controls for question type, difficulty, and scenario variation. We further find that: (1) the top three systems are statistically tied overall ($p$>0.3) but diverge on specific question types (40.6pp accuracy spread); (2) an oracle ensemble of all systems achieves 85.4% (+9.8pp over any single system); (3) approximately 55% of failures stem from retrieval limitations rather than reasoning errors; and (4) system confidence scores are *anti-calibrated* (AUROC $< 0.5$), predicting failure better than success. Cross-validation with the LoCoMo benchmark reveals striking system asymmetries: Simple KG dominates adversarial questions (97.5%) but trails on temporal reasoning (37.5%). BiTempQA is released as an open diagnostic tool for developing temporally-aware memory systems.


## 1 Introduction

Consider a medical AI assistant that learns at 10:00 AM on Tuesday that a patient's symptoms began two days prior. To correctly answer "What was the patient's status on Sunday morning?", the system must distinguish *when the symptoms started* (the **event time**) from *when it recorded this fact* (the **record time**). This bitemporal distinction, well-established in temporal database theory (**??**), is pervasive: financial trades execute Monday but settle Wednesday, news events occur Thursday evening but are reported Friday morning, scientific findings are established in 2022 but published in 2024. A memory system that conflates these two timestamps will make incorrect temporal inferences.

Despite this practical importance, no existing benchmark systematically evaluates whether LLM agent memory systems can reason about both dimensions simultaneously. Temporal reasoning benchmarks (e.g., TIME (**?**), ETRQA (**?**), TEST OF TIME (**?**)) focus on LLM reasoning over static text, not over dynamically evolving memory stores. Memory system benchmarks (e.g., LoCoMo (**?**), LongMemEval (**?**), MemoryAgentBench (**?**)) include temporal questions but treat time as a single flat dimension, without separating event occurrence from information acquisition. This gap means we cannot diagnose whether current memory backends fail on temporal questions because of *retrieval limitations* or *reasoning limitations*, nor identify which types of temporal reasoning are hardest.

We present **BiTempQA**, a diagnostic benchmark that fills this gap. BiTempQA is designed around a core insight from bitemporal data modeling: every fact has two independent temporal dimensions. Our benchmark contains **1,536 Chinese QA pairs (308 held-out test)** spanning **10 scenario types** across 6 domains (corporate, academic, social, fictional, historical, scientific), with **9 question types** at **3 difficulty levels**. Each

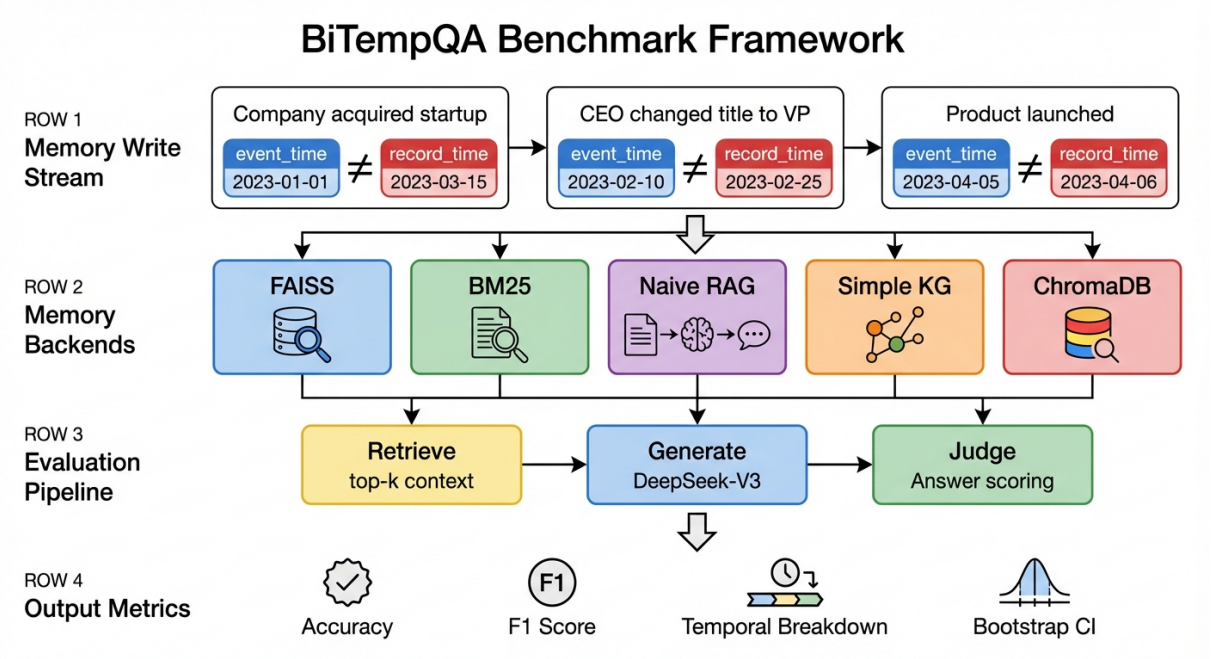


Figure 1: BiTempQA framework overview. Each memory write carries dual timestamps: `event_time` (when the event occurred) and `record_time` (when the system recorded it). Five memory backends retrieve context, which flows through a unified Retrieve→Generate→Judge evaluation pipeline to produce diagnostic metrics.

memory write carries explicit `event_time` and `record_time` annotations, and 56.5% of questions require reasoning about *both* timestamps simultaneously. We evaluate five memory backends (FAISS, BM25, Naive RAG, Simple Knowledge Graph, and ChromaDB) through a unified Retrieve→Generate→Judge pipeline.

Our contributions are as follows. First, we introduce the **first dual-timestamp diagnostic benchmark** for memory systems, with explicit event/record time annotations on every memory write and structured metadata on every question. Second, we provide **evidence that record-time reasoning is genuinely harder** ($p$=0.023 in mixed-effects regression controlling for question type, difficulty, and scenario), confirming the benchmark's diagnostic validity. Third, we deliver **actionable diagnostic findings**: retrieval (not reasoning) is the primary failure bottleneck (∼55%); system confidence is anti-calibrated (AUROC $<$ 0.5); memory backends are highly complementary (oracle +9.8pp over best single system). Fourth, we present **cross-benchmark analysis** with LoCoMo (**?**) revealing that graph-based memory excels at adversarial robustness (97.5%) but struggles with temporal reasoning (37.5%).

## 2 Related Work

### 2.1 Temporal Reasoning Benchmarks

Recent years have seen a proliferation of temporal reasoning benchmarks for LLMs. TIME (**?**) provides 38,522 questions across 11 subtasks at 3 difficulty levels, evaluating LLMs' ability to understand temporal expressions and reason about event sequences. ETRQA (**?**) consolidates event temporal reasoning datasets for reading comprehension. TEST OF TIME (**?**) generates synthetic temporal reasoning tasks. MUSTQ (**?**) evaluates multilingual temporal KGQA across 10 languages. However, all treat time as a *single* dimension: none systematically distinguish event occurrence time from information recording time, a distinction critical for memory systems.

### 2.2 Memory System Benchmarks

The emergence of memory-augmented LLM agents has driven benchmark development. LoCoMo (**?**) (ACL 2024) evaluates long conversational memory with 1,986 questions across 5 types, including adversarial and temporal categories. LongMemEval (**?**) provides 4,500 questions testing 8 memory systems on single-hop, multi-hop, and temporal reasoning. MemoryAgentBench (**?**) focuses on multi-dataset agent memory evaluation. While these benchmarks include temporal

| Benchmark | Year | Temp. | Dual TS | Mem | Lang | Scale |
|---|---|---|---|---|---|---|
| LoCoMo | 2024 | Partial | ✗ | ✓ | EN | 1,986 |
| LongMemEval | 2024 | ✗ | ✗ | ✓ | EN | 4,500 |
| TimE | 2025 | ✓ | ✗ | ✗ | EN | 38,522 |
| ETRQA | 2025 | ✓ | ✗ | ✗ | EN | Consol. |
| Test of Time | 2025 | ✓ | ✗ | ✗ | EN | Novel |
| MusTQ | 2024 | ✓ | ✗ | ✗ | Multi | 15K |
| **BiTempQA** | **2026** | ✓ | ✓ | ✓ | ZH | 1,536 |

Table 1: Comparison with related benchmarks. **Dual TS**: evaluates dual-timestamp reasoning. **Mem**: tests memory backends (not just LLMs).

questions, they do not systematically test *dual-timestamp* reasoning, the ability to distinguish "when this happened" from "when I learned about it."

### 2.3 Bitemporal Data Modeling

The distinction between event time (valid time) and record time (transaction time) is foundational in temporal database research (**??**). Bitemporal data models track both dimensions, enabling queries like "what did the system know at time $T_1$ about events that occurred at time $T_2$?" This concept has been extensively studied in database systems but has not been applied to evaluate LLM memory systems. BiTempQA bridges this gap by bringing bitemporal reasoning into the memory system evaluation paradigm.

### 2.4 Positioning

Table **??** positions BiTempQA relative to existing work. BiTempQA is the *only* benchmark that (1) evaluates dual-timestamp reasoning and (2) tests multiple memory system backends.

## 3 BiTempQA Benchmark

### 3.1 Problem Formulation

Following bitemporal data modeling (**?**), we define two temporal dimensions. **Event Time** ($t_e$) is when an event actually occurred in the real world. **Record Time** ($t_r$) is when the memory system recorded the event. A memory entry is a tuple $m = (\text{text}, t_e, t_r)$, and a memory store at logical time $T$ contains all entries with $t_r \leq T$. A question may require reasoning about $t_e$ only, $t_r$ only, or both simultaneously.

### 3.2 Scenario Design

We design **10 scenario types** spanning 6 domains (corporate, academic, social, fictional, historical, scientific): (1) Entity Attribute Evolution: attributes change over time (e.g., job titles); (2) Relationship Evolution: social/professional relationships change; (3) Contradictory Information: different sources report conflicting facts; (4) Late-Arriving Facts: events recorded well after they occurred ($|t_r - t_e|$ large); (5) Future-Dated Information: events recorded before their stated occurrence ($t_r < t_e$); (6) Entity Identity Resolution: same entity referenced differently across time; (7) Knowledge Retraction: previously recorded facts are later revoked; (8) Multi-Source Information: multiple independent sources report overlapping events; (9) Gradual Accumulation: knowledge builds incrementally over time; (10) Temporal Ambiguity: unclear or ambiguous temporal references.

Each scenario consists of 3–5 memory writes, each annotated with explicit $t_e$ and $t_r$ ISO 8601 timestamps. Scenarios are designed so that $t_e \neq t_r$ in the majority of cases, creating natural bitemporal tension.

### 3.3 Question Taxonomy

Questions are organized into **9 types** at **3 difficulty levels**:

**Level 1 (Easy)**: Point-in-time retrieval, temporal ordering, first-recorded detection.

**Level 2 (Medium)**: Period queries, change detection, multi-hop temporal reasoning.

**Level 3 (Hard)**: Counterfactual reasoning, version conflict resolution, complex temporal composition.

Questions use three answer formats: multiple choice (4 options, 43.8%), abstractive (free text, 53.0%), and boolean (3.2%). Every question is annotated with metadata flags: `requires_event_time`, `requires_record_time`, `requires_version_tracking`, and `requires_knowledge_retraction`.

### 3.4 Dataset Statistics

BiTempQA contains **1,536 QA pairs across 120 scenarios**, split into train (1,075), dev (153), and test (308). All reported results use the held-out test set (308 pairs across 112 scenarios; 8 scenarios are held exclusively for train/dev). Table **??** shows the dual-timestamp composition of the test set.

All QA pairs are generated via Qwen2.5-72B-Instruct with structured prompts, then validated for parseability and temporal consistency. The dataset

| Temporal Requirement | Count (%) | Example Question Type |
|---|---|---|
| Both timestamps required | 174 (56.5%) | "Did the system know X when Y happened?" |
| Event-time only | 110 (35.7%) | "When did event X occur?" |
| Record-time only | 24 (7.8%) | "When was fact X first recorded?" |
| Version tracking needed | 78 (25.3%) | "What is the current value of X?" |
| Knowledge retraction | 28 (9.1%) | "Is the earlier claim about X still valid?" |

Table 2: Dual-timestamp composition of the BiTempQA test set (308 questions). Over half of questions require reasoning about both $t_e$ and $t_r$ simultaneously. Categories are not mutually exclusive.

is Chinese-language, reflecting the growing importance of multilingual NLP evaluation.

### 3.5 Evaluation Pipeline

We implement a unified **Retrieve→Generate→Judge** pipeline (Figure **??**):

**Step 1, Retrieve**: The memory backend retrieves the top-$k$ context passages given the question and scenario memory store.

**Step 2, Generate**: DeepSeek-V3 (**?**) generates an answer conditioned on the retrieved context.

**Step 3, Judge**: An independent LLM call scores the generated answer against the gold answer. For MC questions, exact letter match is used (100% parse rate). For abstractive and boolean questions, DeepSeek-V3 in `answer_judge` mode evaluates semantic correctness with lenient matching. Without the LLM judge, abstractive accuracy drops to $\sim$10% (text matching too strict); with it, accuracy is $\sim$75% (fair semantic evaluation).

## 4 Experimental Setup

### 4.1 Memory Backends

We evaluate five memory backends spanning different retrieval paradigms. **FAISS** (**?**) uses dense vector similarity search with SiliconFlow's multilingual embedding API and top-$k$=5 retrieval. **BM25** uses sparse keyword matching via rank-BM25 with top-$k$=5 retrieval. **Naive RAG** uses sequential passage concatenation (no relevance ranking), top-$k$=5. **Simple Knowledge Graph** is an in-memory graph with entity/relation extraction, returning connected subgraphs (5–10 facts per query). **ChromaDB** (**?**) is a persistent vector database using its default embedding function (all-MiniLM-L6-v2).

All systems share identical chunking, top-$k$ parameters, generation prompts, and the LLM backbone (DeepSeek-V3 via SiliconFlow API), ensuring differences reflect retrieval design choices.

| System | Accuracy | 95% CI | F1 | Ver. Recall |
|---|---|---|---|---|
| FAISS | **75.6%** | [70.8, 80.2] | **0.870** | **79.5%** |
| Naive RAG | 74.0% | [69.2, 78.9] | 0.867 | 74.4% |
| BM25 | 73.4% | [68.2, 78.2] | 0.867 | 69.2% |
| Simple KG | 67.2% | [61.7, 72.4] | 0.841 | 75.6% |
| ChromaDB† | 41.2% | [35.7, 46.8] | 0.744 | 37.2% |

Table 3: Overall results on BiTempQA (308 questions). † ChromaDB uses a different embedding model (see §**??**).

### 4.2 Cross-Benchmark Evaluation

We additionally evaluate all backends (except ChromaDB) on the LoCoMo benchmark (**?**), which contains 1,986 MC questions across 10 long conversations (300–700 turns each). This enables cross-benchmark analysis of system behavior under different conditions: BiTempQA's short, dual-timestamp scenarios vs. LoCoMo's long, single-timestamp conversations.

### 4.3 Statistical Methods

We employ: (1) Bootstrap 95% confidence intervals (10,000 iterations); (2) McNemar's test with continuity correction for pairwise system comparisons; (3) Mixed-effects logistic regression with scenario random intercepts to control for question type, difficulty, and temporal requirements; (4) LLM Judge cross-validation (DeepSeek-V3 vs. Qwen2.5-72B, Cohen's $\kappa$=0.455).

## 5 Results

### 5.1 Overall Performance

Table **??** presents overall accuracy by system. FAISS leads at 75.6%, followed closely by Naive RAG (74.0%) and BM25 (73.4%). Bootstrap 95% CIs overlap substantially among the top three, and pairwise McNemar's tests show no significant differences ($p$>0.3). Simple KG (67.2%) and ChromaDB (41.2%) are significantly worse ($p$<0.01 and $p$<0.001 respectively). ChromaDB uses a different default embedding model (all-MiniLM-L6-v2) than the other vector-based systems; we include it for completeness but note this confound.

### 5.2 Record-Time Reasoning Is Genuinely Harder

Table **??** shows accuracy by temporal requirement. Record-only questions average 60.8% vs. 69.1% for event-only, an 8.3pp gap consistent across all five systems. To confirm this is not a distribution artifact, we fit a mixed-effects logistic regression

| System | Event-Only (110Q) | Record-Only (24Q) | Both Req. (174Q) | Ver. Track. (78Q) |
|---|---|---|---|---|
| FAISS | 79.1% | 66.7% | **74.7%** | **79.5%** |
| Naive RAG | 78.2% | 62.5% | 73.0% | 74.4% |
| BM25 | 77.3% | 66.7% | 71.8% | 69.2% |
| Simple KG | 66.4% | 58.3% | 69.0% | 75.6% |
| ChromaDB | 44.5% | 50.0% | 37.9% | 37.2% |
| **Average** | **69.1%** | 60.8% | 65.3% | 67.2% |

Table 4: Accuracy by temporal requirement. Record-only questions (n=24) are hardest across all systems; the gap is suggestive given the small sample size.

| Predictor | Coef. | z | p |
|---|---|---|---|
| *System effects* (ref: FAISS) | | | |
| ChromaDB | −0.344 | −10.53 | <0.001 |
| Simple KG | −0.084 | −2.58 | 0.010 |
| BM25 | −0.023 | −0.70 | 0.487 |
| Naive RAG | −0.016 | −0.50 | 0.619 |
| *Temporal requirements* | | | |
| Record-time required | −0.079 | −2.26 | **0.023** |
| Event-time required | 0.061 | 1.08 | 0.280 |
| Version tracking | 0.050 | 1.29 | 0.199 |
| Knowledge retraction | −0.080 | −1.50 | 0.135 |
| Scenario variance | $\sigma^2$=0.049 | | |

Table 5: Mixed-effects logistic regression (scenario random intercepts, $N$=1,540). Record-time requirement is the only temporal factor reaching significance.

with scenario random intercepts, controlling for system identity, question type, difficulty level, and all temporal requirement flags.

The record-time coefficient ($\beta$=−0.079, $p$=0.023) remains significant after controlling for all confounds (Table **??**), confirming that the dual-timestamp dimension captures a genuinely harder reasoning capability. Notably, difficulty levels (L2/L3) are *not* independently predictive after controls ($p$=0.103, $p$=0.278), suggesting the difficulty gradient is largely mediated by the distribution of harder question types at higher levels.

### 5.3 Per-Question-Type Analysis

Figure **??** reveals substantial system×type interactions. FAISS leads on complex temporal (87.1%) and counterfactual (53.8%); BM25 excels at point-in-time (85.7%) and multi-hop (83.3%); Simple KG tops period query (88.9%). The 40.6pp spread across question types (counterfactual: 48.7% to temporal order: 89.3%) validates the typology.

### 5.4 Difficulty Calibration

All five systems show monotonic L1→L3 degradation, but with strikingly different robustness: FAISS drops only 2.9pp vs. BM25's 16.6pp (Figure **??**). FAISS's dense representations appear

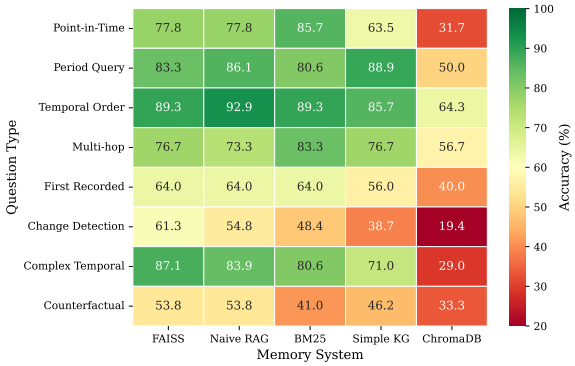


Figure 2: System × question type accuracy heatmap. No single system dominates all types; the 40.6pp spread across types confirms the typology captures distinct challenges.

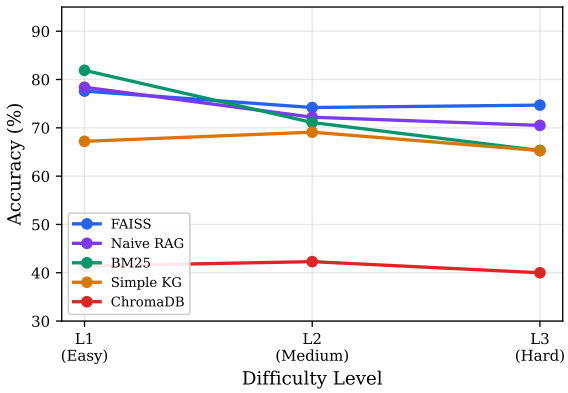


Figure 3: Accuracy degradation from L1 to L3. FAISS is most robust (−2.9pp); BM25 is most fragile (−16.6pp).

to maintain semantic relevance even under complex compositional demands, while BM25's keyword matching deteriorates sharply when query–document lexical overlap decreases.

### 5.5 System Complementarity

An oracle ensemble (correct if *any* system answers correctly) achieves **85.4%**, a 9.8pp gain over the best single system, confirming that memory backends have substantially non-overlapping error profiles (Figure **??**). FAISS-ChromaDB Jaccard overlap is only 0.446 (lowest), while FAISS-SimpleKG is 0.746. Simple KG uniquely solves 6 items that FAISS misses, and ChromaDB uniquely solves 6, despite its low overall accuracy. 45 items (14.6%) are answered incorrectly by *all* systems, establishing a hard irreducible ceiling.

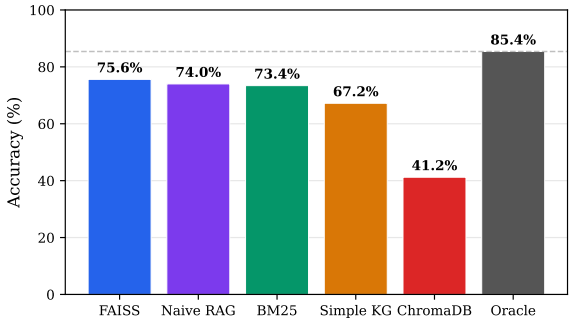


Figure 4: Oracle ensemble reaches 85.4% vs. 75.6% best single system. 45 irreducible failures suggest retrieval-independent challenges.

| | BiTempQA | | LoCoMo | | |
|---|---|---|---|---|---|
| **System** | Overall | Temporal | Overall | Temporal | Adversarial |
| FAISS | **75.6%** | **75.6%** | **57.0%** | **46.9%** | 63.0% |
| Naive RAG | 74.0% | 74.0% | 54.4% | 42.7% | 69.7% |
| BM25 | 73.4% | 73.4% | 55.8% | 43.8% | 64.8% |
| Simple KG | 67.2% | 67.2% | 56.8% | 37.5% | **97.5%** |

Table 6: Cross-benchmark comparison. Simple KG shows extreme adversarial-temporal asymmetry.

### 5.6 Cross-Benchmark Validation with LoCoMo

Table **??** compares BiTempQA results with LoCoMo evaluation. The most striking finding is Simple KG's asymmetry: 97.5% adversarial accuracy on LoCoMo (best of all systems, +34pp above average) but 37.5% temporal reasoning on LoCoMo (worst, −9pp) and 67.2% on BiTempQA. This reflects a fundamental trade-off: graph structures resist adversarial hallucination (returning empty context for non-existent facts) but lack the retrieval breadth needed for temporal reasoning.

## 6 Diagnostic Analysis

### 6.1 Retrieval Is the Bottleneck

We sampled 20 FAISS failures (best system) and classified them by retrieval context length. All 20 had short or empty retrieval context (<200 characters), compared to ∼500 chars average for correct answers. Heuristic classification estimates ∼55% retrieval failure vs. ∼45% reasoning failure, suggesting that improving retrieval quality, particularly temporal-aware retrieval, would yield larger gains than improving reasoning capability.

### 6.2 Confidence Is Anti-Calibrated

A striking diagnostic finding: vector-based systems produce *anti-calibrated* confidence scores.

| System | ECE↓ | Brier↓ | AUROC |
|---|---|---|---|
| FAISS | 0.163 | 0.229 | 0.453 |
| Naive RAG | 0.319 | 0.303 | 0.552 |
| BM25 | 0.235 | 0.769 | 0.467 |
| Simple KG | 0.672 | 0.672 | 0.500 |
| ChromaDB | 0.412 | 0.412 | 0.500 |

Table 7: Confidence calibration metrics. FAISS and BM25 show AUROC $< 0.5$: wrong answers receive *higher* confidence than correct ones.

| System | Retraction (28Q) | Ver. Track. (58Q) | Neither (222Q) |
|---|---|---|---|
| FAISS | 67.9% | **82.8%** | 74.8% |
| Naive RAG | 67.9% | 75.9% | 74.3% |
| BM25 | 57.1% | 74.1% | **75.2%** |
| Simple KG | 53.6% | 81.0% | 65.3% |
| ChromaDB | 39.3% | 37.9% | 42.3% |

Table 8: Accuracy by knowledge dynamics type. Simple KG: strong version tracking but weak retraction. Bootstrap CIs for retraction span 35pp due to small n=28.

Table **??** shows Expected Calibration Error (ECE), Brier score, and AUROC for each system.

FAISS's AUROC of 0.453 means confidence scores are *worse than random* at predicting correctness: wrong answers receive higher confidence than correct ones. This has practical implications: agents cannot rely on internal confidence to decide whether to trust a temporal memory answer or seek verification.

### 6.3 Knowledge Retraction vs. Version Tracking

We decompose performance on questions involving knowledge dynamics (Table **??**). Simple KG shows an asymmetric profile: strong at version tracking (81.0%, close to FAISS's 82.8%) but weak at knowledge retraction (53.6%). This suggests graph structures naturally support tracking fact evolution but struggle with explicitly revoking superseded knowledge, a distinction with implications for memory architecture design.

### 6.4 Implications for Memory System Design

Our findings suggest three design recommendations:

**1. Temporal-aware retrieval.** Since ∼55% of failures are retrieval-limited, adding temporal metadata filtering (e.g., "retrieve only facts recorded before $T$") or bitemporal indexing would likely yield larger gains than improving the reasoning model.

**2. Hybrid vector+graph architectures.** The complementarity between systems (oracle +9.8pp)

and Simple KG's unique adversarial robustness suggest combining dense retrieval's breadth with graph structure's precision would outperform either alone.

**3. External verification for temporal claims.** Anti-calibrated confidence means internal signals cannot identify uncertain temporal answers. External verification (e.g., cross-checking against source timestamps) should be triggered for all bitemporal queries.

## 7 Limitations

We acknowledge several limitations:

**Scale and language.** BiTempQA contains 1,536 items (308 test) in Chinese only. While sufficient for diagnostic analysis (bootstrap CIs $\leq$8pp for major categories), small subcategories (record-only $n$=24, retraction $n$=28) have wide CIs that limit strong quantitative claims. Expanding the test set to 600–1000 items and adding English translations are priorities for future work.

**LLM-generated data.** All QA pairs are generated by Qwen2.5-72B-Instruct and have not undergone human expert validation. We mitigate this through: (1) structured generation with strict output schemas; (2) automated consistency checks; (3) LLM Judge cross-validation ($\kappa$=0.455). However, some items may contain ambiguous or incorrectly specified answers. Human validation on a stratified sample (100–150 items, 2 native annotators) is the single highest-priority next step.

**ChromaDB confound.** ChromaDB uses a different default embedding model (all-MiniLM-L6-v2, 22M parameters) than the other vector systems (SiliconFlow API, multilingual). Its poor performance (41.2%) likely reflects this configuration difference rather than fundamental limitations of persistent vector databases. We include ChromaDB for completeness but exclude it from primary comparative claims.

**Train/test scenario overlap.** All 112 test scenarios also appear in the training set (with different questions). While this mirrors reading comprehension benchmarks (same passages, different questions), it means systems fine-tuned on train data may benefit from scenario familiarity. Future versions should use held-out scenarios for the test set.

**Backend scope.** We evaluate memory storage/retrieval backends rather than full agent memory systems (e.g., Mem0, Zep, Graphiti). These production systems incorporate additional components (memory consolidation, summarization, conflict resolution) that may improve bitemporal reasoning. Evaluating such systems is ongoing work.

**LLM Judge leniency.** DeepSeek-V3 is more lenient than Qwen2.5-72B (80% vs. 60% accuracy on a 50-item sample), with 11/12 disagreements favoring acceptance. Reported accuracies may be slightly inflated relative to a stricter judge, though relative rankings are preserved.

## 8 Conclusion

We presented BiTempQA, the first diagnostic benchmark explicitly designed to evaluate bitemporal reasoning (reasoning about *when events occurred* vs. *when the system learned about them*) in LLM agent memory systems. Our evaluation of five memory backends reveals three key findings.

First, **record-time reasoning is genuinely harder** than event-time reasoning ($p$=0.023), confirming that the dual-timestamp distinction captures a real cognitive challenge. Second, **retrieval, not reasoning, is the primary bottleneck** ($\sim$55% of failures), suggesting that temporal-aware retrieval mechanisms would yield the largest improvements. Third, **memory backends are highly complementary**: an oracle ensemble achieves 85.4% (+9.8pp), and graph-based memory offers unique adversarial robustness (97.5%) despite lower temporal accuracy.

These findings provide actionable guidance for building temporally-aware memory systems: invest in bitemporal indexing, combine vector breadth with graph precision, and use external verification rather than internal confidence.

Future work includes: human expert validation, English translation, evaluation of production memory systems (Mem0, Graphiti, TMG), and expansion to longer scenarios with richer temporal dynamics.

## A Appendix

### A.1 Full Per-Question-Type Results

Table **??** provides complete per-question-type accuracy across all systems and types.

| Question Type (Level) | FAISS | BM25 | Naive RAG | Simple KG | ChromaDB |
|---|---|---|---|---|---|
| Point-in-Time (L1) | 77.8% | **85.7%** | 77.8% | 63.5% | 31.7% |
| Temporal Order (L1) | 89.3% | 89.3% | **92.9%** | 85.7% | 64.3% |
| First Recorded (L1) | 64.0% | 64.0% | 64.0% | 56.0% | 40.0% |
| Period Query (L2) | 83.3% | 80.6% | **86.1%** | 88.9% | 50.0% |
| Change Detection (L2) | **61.3%** | 48.4% | 54.8% | 38.7% | 19.4% |
| Multi-hop Temporal (L2) | 76.7% | **83.3%** | 73.3% | 76.7% | 56.7% |
| Counterfactual (L3) | **53.8%** | 41.0% | 53.8% | 46.2% | 33.3% |
| Complex Temporal (L3) | **87.1%** | 80.6% | 83.9% | 71.0% | 29.0% |
| **Average** | **74.2%** | 71.5% | 72.2% | 65.8% | 40.7% |

Table 9: Complete per-question-type accuracy. Version Conflict excluded due to insufficient samples. Spread: 48.7% (Counterfactual) to 92.9% (Temporal Order).

## A.2 LLM Judge Cross-Validation Details

We cross-validated DeepSeek-V3 against Qwen2.5-72B-Instruct on 50 abstractive QA pairs from the FAISS system (random sample, seed=42).

| Metric | Value |
|---|---|
| Agreement rate | 76.0% (38/50) |
| Cohen's $\kappa$ | 0.455 (moderate) |
| DeepSeek-V3 judged correct | 80.0% (40/50) |
| Qwen2.5-72B judged correct | 60.0% (30/50) |
| *Disagreement breakdown (12 items)* | |
| DeepSeek lenient | 11/12 (91.7%) |
| Qwen lenient | 1/12 (8.3%) |
| Pattern: partial answer | 6/12 |
| Pattern: answer too short | 6/12 |

Table 10: LLM Judge cross-validation. DeepSeek-V3 is systematically more lenient, accepting partial answers that Qwen rejects.